

The Design and Research of a War Game Simulation System for Equipment Support

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Abstract—To meet the evolving demands of equipment support in future warfare scenarios, the utilization of war games within the realm of equipment support is being expedited to effectively enhance the efficiency of equipment support. On the basis of explaining the concept of the equipment support war game simulation system, this paper makes an in-depth analysis from three aspects: military requirement, application requirement and system function requirement. Furthermore, it proposes an overall structure for this system based on the implementation process of equipment support. The research findings can serve as a theoretical reference for both the development and practical application of future equipment support war game systems.

Keywords—war game simulation; equipment support; system design

I. INTRODUCTION

The war game has evolved over hundreds of years, progressing from the traditional manual turn-based system to today's computer dynamic real-time system. Its innovation and development have always been highly valued by military strategists throughout history [1]. In recent years, the world's military powers have vigorously developed war games with the assistance of high-tech tools such as artificial intelligence and computers. China has also conducted research and development on war games, achieving significant results in realizing the dynamic function of combat confrontation. However, most of the war game systems developed at this stage are biased towards combat, neglecting equipment support activities that maintain and restore military combat effectiveness and significantly impact the progress and outcome of wars. This lack leads to a decline in the credibility of deduction results. With the increasing application of information and intelligent equipment in combat, modern warfare is becoming increasingly reliant on equipment support. Therefore, the study of military game calculation in equipment support holds significant practical importance.

II. THE CONCEPT AND CONNOTATION OF WAR GAMES

A war game refers to a tool that utilizes simulation technology to construct a realistic battlefield environment. It involves the design of maps and chess pieces according to specific rules, as well as the establishment of action probability tables that reflect the uncertainty of the battlefield based on data accumulated from historical experience and training. Furthermore, dynamic methods are employed to present simulated war scenarios for all parties involved in order to conduct simulated deduction and analysis research [2]. With the development of technology, war games have evolved from manual war games to computer war games, and have now

developed into an information-based war game simulation system integrating deduction, operation research analysis, and equipment simulation (as shown in Figure 1).

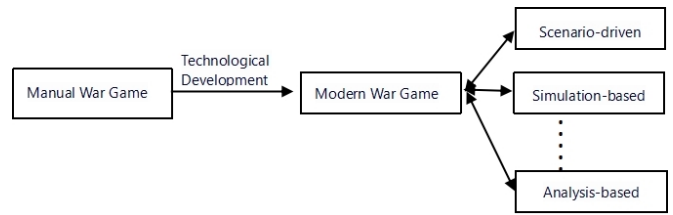


Figure 1. Development and evolution of war games.

The equipment support war game refers to the process in which participants use chessboards and chess pieces representing the battlefield and its military forces, or computer-based virtual battlefield environments and troops, to simulate wartime equipment support activities, conduct process deduction and evaluation optimization of support plans, based on relevant rules and principles of probability theory. It is the application of war game principles in the field of equipment support [3]. The equipment support war game system is a simulation method for equipment support activities, set against the backdrop of modern warfare. It is driven by support missions and follows the basic principles of war gaming. This system aligns with the equipment support processes and command relationships, corresponds to the equipment system, and matches the operational war game system. It can effectively guide the innovation of equipment support theories, support drills, and support effectiveness evaluation. Its core lies in grasping dynamic laws, enhancing comprehensive support capabilities, and gaining a competitive advantage for victory.

III. EQUIPMENT SUPPORT WAR GAME SIMULATION SYSTEM DEMAND ANALYSIS

A. Military demand analysis

The ultimate goal of equipment support activities is to serve combat operations, and the level of equipment support capabilities is directly related to the success or failure of combat actions [4].

(1) There is a need to effectively enhance the comprehensive capabilities of support commanders. The complexity and uncertainty of modern warfare environments present unprecedented challenges to equipment support work [5], placing greater demands on the strategic vision, decision-making abilities, and organizational and coordination capabilities of equipment support commanders. It is imperative for equipment support commanders to acquire advanced

support technologies and methods, familiarize themselves with the performance and operational requirements of various types of equipment, master complex support processes, and response strategies, as well as enhance their keen insight and judgment of battlefield situations by simulating real battlefield environments and support processes. This will ultimately improve decision-making efficiency and accuracy.

(2) There is a need to adapt to future warfare equipment support. Contemporary warfare necessitates the military to possess adaptable and multifaceted support capabilities in order to accommodate various battlefield environments and operational support requirements [6]. The war game simulation system simulates a wide range of future combat scenarios and unexpected situations and challenges that may arise, predicting the support requirements of equipment under complex conditions. This provides commanders with an important basis for formulating response strategies and optimizing equipment support plans in advance, ensuring that equipment support can play a critical role in future wars.

(3) There is a need to make up for shortcomings in the actual combat training of equipment support. Currently, the mechanism for actual combat training of equipment support with real soldiers and equipment is sound, with diverse methods and means. However, such training requires a significant amount of manpower, material resources, and financial resources. It is also time-sensitive and comes with potential safety hazards, making it challenging to achieve normalization. The equipment support department utilizes the equipment support war game simulation system to create a highly realistic virtual battlefield environment. This system simulates various complex battlefield conditions and equipment support requirements, allowing participants to efficiently and safely conduct normalized training without actual equipment.

B. Application needs analysis

The equipment support war game simulation is a rehearsal process that occurs throughout the entire support operation. It follows the support procedures and combats progress to practice work deployments and support activities in each phase of the support operation [7]. The application demand model of the equipment support war game deduction system (as shown in Figure 2) should aim to enhance the support command capability under intelligent conditions, reflecting the characteristics of model network support, resource sharing, element linkage, and effective command. It should highlight the full-process simulation of equipment support activities, realize the interconnection with the combat deduction system, and effectively integrate with the equipment system.

(1) Real-time capability. The rapid changes in the war environment necessitate the system to promptly reflect the latest battlefield dynamics. The system must update various data, such as force distribution, equipment status, and battlefield environment in real-time to ensure that equipment support personnel can access the most accurate information immediately. The need for real-time capability is not only evident in data updates but also encompasses the system's instantaneous analysis and prediction of the battlefield situation. Through efficient algorithms and powerful computing capabilities, the system can precisely analyze the battlefield

situation in a very short time, providing timely and effective decision support for support activities.

(2) Flexibility. The war gaming system must be capable of accommodating a wide range of simulation scenarios, encompassing diverse battlefield environments, support objectives, and force configurations. Users should have the ability to select appropriate simulation scenarios based on specific needs and customize simulation rules and parameters to meet varying requirements, thereby enhancing the targeted nature of training. Furthermore, the system should be able to interface effectively with other simulation systems by adopting standard formats. Additionally, it should allow for continuous upgrades and improvements in line with technological advancements and evolving military demands.

(3) Interactivity. The system must offer an intuitive and user-friendly interface and operation mode, as well as rich data visualization tools, to enable users to conveniently and intuitively set simulation parameters, adjust force configurations, and observe simulation results. Furthermore, the system should support multiple users in simultaneous online simulations and collaborations, facilitating communication and discussion among them. This will allow users to gain a more realistic experience during the simulation process, thereby enhancing the effectiveness of the simulation.

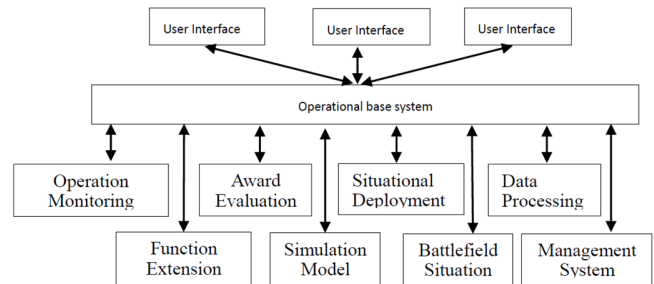


Figure 2. The application demand model for equipment support wargame deduction system.

C. Analysis of system functional requirements

The equipment support war gaming system should enable personnel to analyze and predict equipment support requirements based on simulated battlefield environments. By following the support processes and relationships, they can issue relevant support orders and instructions using game pieces, as well as evaluate the simulation process and results (as shown in Figure 3).

(1) Actual combat simulation and training function. By creating a highly realistic battlefield environment, trainees can experience authentic combat situations in a virtual space. This not only simulates complex tactics and battle rules, providing efficient and cost-effective training methods but also offers real-time feedback on the deduction results based on the trainees' decisions and operations, helping them to become familiar with the support process and improve their command capabilities.

(2) Data analysis function. In modern warfare equipment support, data has become a crucial foundation for decision-making [8]. The system must facilitate the comprehensive collection of fundamental data such as terrain, equipment,

deployment, and operational data including commands, instructions, and support processes throughout the entire process. It should also accommodate various collection methods such as manual input and automatic recording, store data in a standardized format, and standardize it to enable quick and flexible access while effectively supporting rehearsal and review. During the rehearsal process, the system needs to dynamically present information on troop deployment for both enemy and friendly forces, firepower allocation, battle damage assessment, and equipment requirements. It should realize visualization of the battlefield situation to assist commanders in real-time understanding of battlefield dynamics. Simultaneously, the system needs to utilize advanced algorithms to collect and analyze battlefield data in order to predict support needs and future development trends. This will enhance commanders' perception of the battlefield situation while providing a solid basis for command support decisions.

(3) Guidance and control function. The deduction activity is not conducted continuously from start to finish. The adjudicator must establish appropriate time frames for different aspects of the deduction process and provide guidance and oversight as needed throughout the process. Both the red and blue sides are responsible for managing the supply and maintenance activities of equipment and ammunition in accordance with support requirements, taking into account support conditions and the

environment, in order to ensure the orderly progression of the deduction process.

(4) Assisting in decision-making. Commanders must utilize the war game system's assisting decision-making function to forecast battle losses and requirements, provide real-time feedback on the impact of safeguard decisions, comprehend the strengths and weaknesses of different plans, continuously optimize safeguard plans, and enhance the scientificity, accuracy, and efficiency of decision-making during the deduction process.

(5) Adjudication and evaluation function. The system can accurately reflect the level of support effectiveness through quantitative analysis based on simulated support data, in various aspects such as support plans and equipment support effects. This assists commanders and trainers in analyzing existing problems. The system should evaluate both the overall deduction and the support activities; it not only supports real-time evaluation but also allows for comparative analysis of multiple simulation results, thereby discovering patterns and providing strong support for optimization and decision-making [9]. At the same time, this function can also provide a scientific basis for equipment procurement and force allocation, ensuring the rational allocation and efficient utilization of military resources [10].

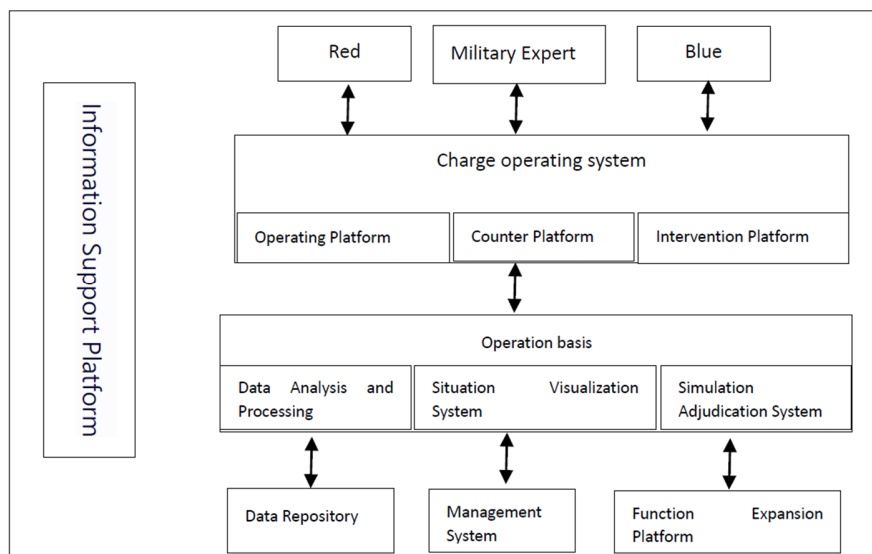


Figure 3. Functional requirement model for equipment support wargame deduction system.

IV. DESIGN OF EQUIPMENT SUPPORT WAR GAME SIMULATION SYSTEM ARCHITECTURE

A. Design principles

The diverse design purposes of various war games necessitate the implementation of distinct design principles, which in turn cater to a wide range of participants [11]. Therefore, the design of equipment that supports the war game deduction system, based on traditional war game design, fully leverages modern technologies and methods. It combines the purpose of system design, expands and innovates methods and contents, and follows the principles of authenticity, intelligence,

and security to ensure the effectiveness and practicality of the system.

(1) The principle of authenticity. The system must be compatible with operations and in accordance with the equipment support process and support relationship, to simulate the equipment support process in the actual combat environment as authentically as possible. This will provide accurate and reliable deduction results.

(2) The principle of intelligence. The system is required to utilize advanced technologies for intelligent analysis, prediction, and auxiliary decision-making. This will help users

to more efficiently and accurately respond to complex and ever-changing equipment support environments.

(3) The principle of security. The equipment support war game deduction system involves a large amount of sensitive military data, therefore, strict security measures must be taken, including user access permissions, data encryption, operational control, etc., to ensure the security and confidentiality of data as well as the stable operation of the system.

B. Overall system architecture model

The equipment support war game is built upon the operational war game and focuses on researching the framework of the equipment support war game deduction system by inheriting elements from the operational war game [12]. The construction of an equipment support war game deduction system architecture model (as shown in Figure 4) involves the subdivision of five sub-system models under the overarching system architecture. This approach enables a clearer delineation of functions and responsibilities among the various components while facilitating better coordination among them. As a result, the overall performance, fluency, and stability of the system's operation are ensured.

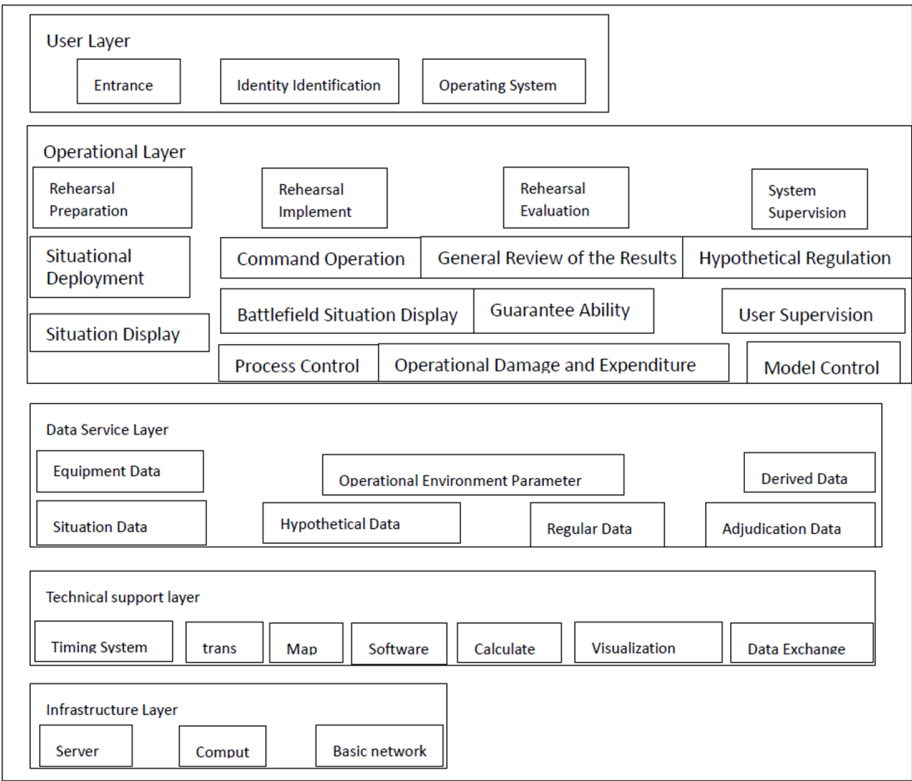


Figure 4. Architecture model of equipment support wargame deduction system.

1) *User interaction layer model.* The User Interaction Layer plays a crucial role in directly communicating and interacting with users. As the face of the system, the design and implementation of the user interaction layer model directly impact the user experience and satisfaction. Firstly, the operation interface is intuitive and concise. Initially, four interface styles are designed, with personalized settings for buttons and icons in each mode. Participants are randomly selected in a ratio of 50% for general staff, 35% for middle managers, and 15% for senior leaders. After multiple rounds of testing, the results show that 90% of the participants prefer the intuitive and concise style. Secondly, the operation mode is simple and easy to learn. Users can complete tasks such as command input, information inquiry, and result viewing by simply clicking, dragging, and keyboard input, enabling

interaction with the system. The system supports up to 180 users working simultaneously. Lastly, the User Interaction Layer seamlessly collaborates with other layers of the system by accurately transmitting user instructions and requirements to the Operation Layer for processing while presenting processing results to users in a visual manner. Meanwhile, it maintains close contact with the Data Service Layer to ensure real-time data updates.

2) *Operational layer model.* The Operation Layer is responsible for implementing deduction logic, processing user requests, and adjudicating deduction results. The design and implementation of this layer directly determine the system's deduction capabilities, accuracy, and reliability. Firstly, the operation layer realizes various deductions based on the rules of equipment support war game, including but not limited to

calculating parameters such as mobility, combat effectiveness, and support capacity according to the attributes of the game pieces and the terrain environment, updating the status of the game pieces based on the combat process, and triggering corresponding adjudication logic according to specific events[13]. During the test, the commander issues an instruction of "simultaneous support in four different scenarios," with 80 personnel working simultaneously. The system completes user demand analysis within 30 seconds and generates corresponding parameters and intentions. Throughout the process, the system operates smoothly with a high degree of achievement. Secondly, the Operation Layer establishes a close collaboration with the Data Service Layer by retrieving necessary data such as map data, equipment basic data, piece data, and rule data for deduction calculations[14]. It also synchronizes data changes generated during the deduction process to ensure timeliness and consistency of data.

3) *Data service layer model.* The Data Service Layer is responsible for data storage, retrieval, and management, as well as ensuring data integrity, consistency, and security. First of all, the data service layer is to provide various data resources. The data service model enters more than 6470 pieces of basic equipment data and models in 13 major categories and 1241 sub-categories, more than 108 types of map scene data, 113 war cases, 5 types, and 87 pieces (sets) of simulation weapons, as well as operational support data, dynamic data during the simulation process, etc. These data are the basis for deduction activities and the basis for autonomous learning. In addition to managing the storage of large-scale data efficiently, the Data Service Layer also ensures data integrity and consistency in real-time during the deduction process. It continuously reflects changes in battlefield situations, movement of pieces, and depletion or damage of equipment in the stored data while maintaining consistency among them. To prevent potential loss or corruption of important information due to unforeseen circumstances or unauthorized access attempts by malicious entities, backup functions are incorporated into this layer's design which allows for recovery if necessary. Last but not most importantly, given that military sensitive information is involved in war game deductions, encryption processes are implemented within this layer to ensure that secure access control measures are applied, so only authorized users can access relevant information preventing any unauthorized modifications or leakage of sensitive information from occurring.

4) *Technology support layer.* The technology support layer serves as the foundation of the equipment support war game simulation system architecture, providing stable and reliable technical support and operating environment for the entire system. It encompasses various aspects such as hardware, networks, and underlying software required by the system to ensure its normal operation and high-efficiency performance. Firstly, the technology support layer provides the hardware and network resources necessary for the system. These resources

include high-performance servers, storage devices, network switches, etc., which collectively constitute a stable and reliable hardware platform offering ample computing power and data storage capacity for the system. Meanwhile, the technology support layer is also responsible for network planning, configuration, and maintenance to ensure smooth communication between various components within the system. Secondly, the technology support layer oversees the management and maintenance of the system. It employs advanced management systems and technologies to ensure the effective operation of simulations. Additionally, it provides software support and development tools including operating systems, software technologies optimization algorithms, and development tools that offer the necessary support environment for system development operation[15]. By continuously introducing new technologies and updating optimizing system software, the technology supports stability and high-efficiency performance of the whole system.

5) *Infrastructure layer.* The infrastructure layer plays a crucial role as the foundation of the equipment support war game simulation system, responsible for ensuring the stable and efficient operation of the system. It serves as a solid base for the entire system, encompassing physical devices such as servers, storage devices, switches, and a network environment that meets system requirements. Additionally, it includes security policies such as security management systems and operational norms. Constructing this layer effectively is essential to providing reliable support for the entire system, ensuring its stability, efficiency, and secure operation.

V. KEY TECHNOLOGIES

A. Modeling and simulation technology

Modeling and simulation technology serve as the cornerstone of the equipment support war game deduction system. Firstly, a comprehensive and exhaustive analysis of the structure, performance, and functions of the equipment is conducted, and a large amount of experimental data and performance parameters are collected and integrated to provide accurate resources for the construction of mathematical models. Secondly, through simulation technology, these mathematical models are transformed into vivid virtual battlefields, simulating the complex interaction process between equipment, troops, terrain, and weather. Experimental data shows that modeling and simulation can highly realistically reflect the behavioral characteristics in battlefield environments, providing extremely reliable foundations for everything from collaborative combat at the tactical level to decision-making at the strategic level. [15].

B. Data fusion and processing technology

Data fusion and processing technology utilizes advanced data analysis algorithms to integrate data from diverse sources and formats. This integrated data undergoes thorough mining and analysis to extract valuable information, providing comprehensive data support for the system. In order to ensure data accuracy and consistency, preprocessing, cleaning, and transformation operations are essential.

C. Prediction and evaluation technology

Prediction and evaluation technology utilizes historical data, current equipment status, and battlefield environment information to forecast and assess its current state, future trends, reliability, and operational effectiveness using algorithms such as statistics and machine learning [16]. This enables equipment support personnel to anticipate support needs and process risks in advance, providing a scientific basis for formulating targeted support measures and references for equipment support decision-making. Ultimately, this technology significantly enhances the accuracy and effectiveness of decision-making in the field of equipment support.

D. System security and reliability technology

In order to effectively address various security threats and failure challenges, the system utilizes technical measures such as data encryption, access control, and firewalls to ensure the security and confidentiality of system data. Furthermore, it incorporates redundancy design, fault-tolerant technology, and backup and recovery measures to enhance the reliability and stability of the system [17].

VI. CONCLUSION

The equipment support war game deduction system can better adapt to the iterative updating of weapons and equipment and the transformation of training modes, better train the command, coordination, decision-making, analysis, and judgment abilities of support personnel, and improve the comprehensive effectiveness of equipment support. This article only discusses the requirements, framework, and related technologies of the equipment support war game deduction system, and does not fully develop and apply the system. Moving forward, further research will be conducted to study the equipment system, analyze and summarize equipment support cases, formulate scientific simulation rules, and optimize the simulation process, with an aim to achieve the development and application of the equipment support war game simulation system as soon as possible.

REFERENCES

- [1] War Gaming is the "Magician" of Directing Warfare: An Interview with Major General Luo Yuan [EB/OL]. [2008-03-27]. http://news.xinhuanet.com/mil/2008-03/27/content_7866864_1.htm.
- [2] He Changqi. Analysis of Related Concepts and Connotations of War Gaming [J]. Military Digest, 2024(3): 59-62.
- [3] Wang Guiqi, Du Xiaoming, Gu Chuang, Liu Bin, Sun Xiao. Research on the Framework of Equipment Support War Gaming Simulation System [J]. Journal of Equipment Academy, 2012(04): 63-66.
- [4] Wang Hongjun, Zheng Shiming, Zhang Yongliang. Requirements Analysis of Army Command and Training Information System [J]. Command Control & Simulation, 2016, 38(04): 22-27.
- [5] Jiang Bin, Liang Xiao'an, Tang Jing. Multi-Attribute Decision-Making of Equipment Support Plan Based on Effectiveness Evaluation [J]. Fire Command and Control, 2020, 45(11): 88-93.
- [6] Zan Xiang, Chen Chunliang, Zhang Shixin, et al. Research on Task Allocation and Scheduling Framework of Digital Troop Equipment Maintenance Support [J]. Computer Measurement & Control, 2017, 25(9): 99-114.
- [7] Yan Ke, Cai Ya. Technology and Methods of Weaponry Deduction for Army Weaponry System Operation [M]. Beijing: Military Science Press, 2012: 21-25.
- [8] Ye Limin, Gong Li, Liu Zhong. Research on the Design and Modeling of Wargaming Simulation System [J]. Computer & Digital Engineering, 2011, 39(12): 58-56.
- [9] Peng Chunguang, Ju Rusheng, Yang Jianchi, et al. Analysis of Modern Wargaming Technology [J]. Journal of System Simulation, 2009, 21(S2): 97-100.
- [10] Peng Chunguang, Zhao Xinye, Liu Baohong, et al. Overview of Wargaming Technology [J]. System Simulation Technology and Its Application, 2009(11): 366-370.
- [11] Dong Daoyuan, Gao Wei. DARPA Builds Cross-Domain Heterogeneous Battlefield Network to Support "Mosaic Warfare" [J]. Industrial Information Security, 2022, (11).
- [12] Cui Shuang, Li Baogang, Dong Depeng, Wang Tingting, Chen Shengwei. Research on Equipment Support Effectiveness Evaluation Method Based on Wargaming Simulation System [J]. Ship Electronic Engineering, 2023, (8).
- [13] Guo Hongyu, Chu Yang, Liu Zhi, Zhou Yufang. Research on Command Decision-Making in Submarine Offensive and Defensive Confrontation Training Based on Deep Reinforcement Learning [J]. Command Control & Simulation, 2022, (1).
- [14] Zhang Tao, Xiang Qi, Zheng Wanwen, Sun Yuxiang, Zhou Xianzhong. Application of Improved A* Algorithm in Path Planning of Naval Wargaming Simulation [J]. Acta Armamentarii, 2022, (4).
- [15] Zhang Han, Huang Yanyan, Geng Ze, Zhang Zhen. Evaluation Method of Emergency Support Plan Simulation and Deduction in Barracks [J]. Systems Engineering and Electronics, 2022, (11).
- [16] Xiong Xianwei, Chen Lin, Wang Wenzhong. Research on Dynamic Evaluation Method of Equipment Support Wargaming [J]. Fire Control & Command Control, 2021, (1).
- [17] Han Lulu. Construction of Application Ideas of Artificial Intelligence in Computer Wargaming [J]. Software, 2021, (8).